

# Neural Networks Project – Assignment

## Overview

In this assignment, you will design and implement a generic Neural Network (NN) library. Your library must be reusable, extendable, and independent of any specific use case. After building the library, you will choose a real-world problem that can be solved using neural networks and apply your library to implement a complete solution.

The deadline of this phase will be announced and shown on Google Classroom, so it is **your responsibility to check it and remember it.**

## Part 1 – Neural Network Library Requirements

### 1. Architecture & Core Components

- Implement a modular feedforward neural network framework.
- Support multiple layers with configurable number of neurons.
- Implement at least four activation functions: Sigmoid, ReLU, Tanh, Linear.
- Support at least two weight initialization methods (e.g., Random Uniform, Xavier/He).

### 2. Training the Neural Network

- Implement forward propagation.
- Implement full backward propagation using chain rules.
- Support adjustable hyperparameters: learning rate, epochs, batch size.
- Track training loss across epochs.

### 3. Loss Functions

- Implement at least two loss functions: MSE and Cross-Entropy.
- Provide a clean pipeline for training and evaluation.
- Implement prediction for single inputs and batches.
- Expose intermediate values for debugging.

### 6. Hyperparameter Configuration

- Allow full customization of number of layers, neurons, activations, optimizers, learning rate, initialization method, batch size, and epochs.
- Provide default values for all hyperparameters.

### 7. Data Handling

- Support training/testing splitting.
- Provide normalization utilities.
- Handle invalid or missing inputs gracefully.

### 8. Code Quality

- Follow clean code principles and proper software architecture.
- Organize classes and modules by responsibility.
- Avoid hardcoding problem-specific logic inside the library.

## Part 2 – Case Study Using Your Library

You must select (and get approval for) a real-world problem solvable using neural networks.

### **For your case study, provide:**

1. Problem description and rationale.
2. Dataset description and preprocessing steps.
3. Neural network architecture choice and justification.
4. Final training and evaluation results (loss curves, accuracy, etc.).
5. Clear explanation of how your library was used.
6. Graphs, tables, or screenshots supporting your results.

### **Submission Guidelines**

- Submit one ZIP file named: ID1\_ID2\_ID3\_Major\_Group.zip
- Include:
  - Source code
  - Case study report
  - Dataset or link
  - A runnable demo script

# Grading Scheme (100 Marks)

- **Code Structure (10 marks)**
  - Clean code (5)
  - Modular architecture (5)
- **Neural Network Implementation (45 marks)**
  - Forward propagation (15)
  - Backpropagation + training (15)
  - Prediction + evaluation + loss functions (15)
- **Case Study (20 marks)**
  - Problem selection and justification (5)
  - Neural network architecture selection (5)
  - Problem-dependent decisions (10)
- **Demo (25 marks)**
  - Team coordination and coverage (10)
  - Understanding of NN concepts (10)
  - Runtime performance (5)